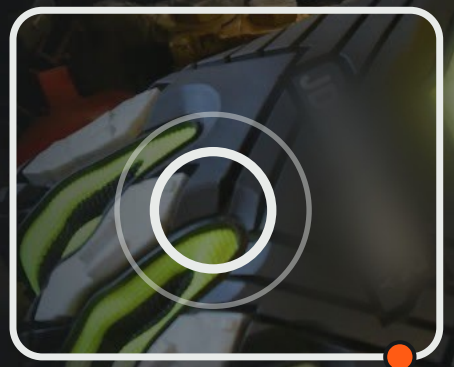
praxis

STEP 4/9

Instructions that can finally see how you're doing

An AI coach for new maintenance technicians. It watches the work through a first-person camera and says the next step.



Seat the clamp past the hose bead

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

CONCEPT MOCK · STOCK PHOTO

THE PROBLEM

A new maintenance technician needs about a year of a senior tech's time to work alone

>1 year

of on-the-job training, with a senior tech as the teacher.

$11–23k

of senior-tech time per new hire, by our estimate.

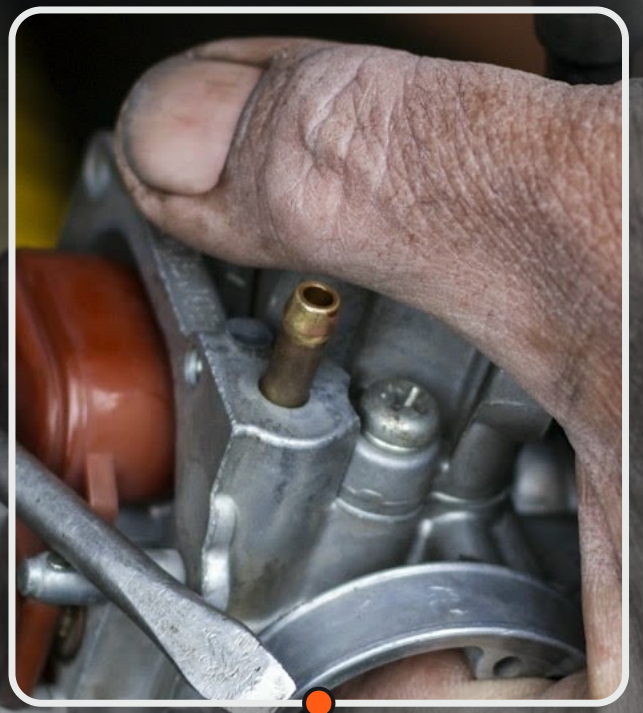
~255,000 openings a year for US maintenance and repair techs, and every new hire pulls a senior tech off the floor.

BLS Occupational Outlook Handbook · BLS Employer Costs for Employee Compensation, June 2025 · estimate is ours

THE PRODUCT

Praxis watches the work and coaches one step at a time

LOAD	the manuals and checklists a shop already has
COACH	on a phone or glasses, one short line at a time
VERIFY	critical steps and keep the record



Loosen the four float-bowl screws a half turn each

CONCEPT MOCK · STOCK PHOTO

HOW FAR ALONG

It already works on real first-person video. The bottleneck is maintenance data

BUILT · OCT 2026

A tracking engine that knows which step of the procedure you're on, and speaks only when it's confident

A live app that runs in real time on a laptop, about 50 ms per frame

An honest test: 48 held-out clips, 235 minutes, with labels we didn't write

WORKS · MEASURED

54% vs 48%

step accuracy, against the same frontier model called every 5 s

1/5 the cost

$0.018 vs $0.085 per minute of video

73%

of clips: it named the job by itself, no setup (median 40 s)

0

times confidently wrong on blurred or blocked views. It asked for a better view instead

THE BOTTLENECK

Every number is from public cooking video. We couldn't get labeled maintenance video.

Mistake alerts are weak: 4 of 17 named a real error.

It hasn't run on glasses yet.

Labeled maintenance recordings and a pair of glasses fix all three. That's the $10k.

48 held-out clips of public first-person video (CaptainCook4D) · one run per system · not yet measured on maintenance

WHY NOW

This year, frontline teams are choosing what replaces their last generation of tools

28 MAY 2026

Autodesk buys MaintainX for $3.6B

16 JUN 2026

Viture + NVIDIA announce coaching glasses

SEPT 2026

Meta opens its glasses' camera to apps

31 DEC 2026

Microsoft ends Dynamics 365 Guides

Q1 2027

Helix and Android XR display glasses ship

$1 an hour

what it costs us to run Praxis today: $0.018 per minute of video

~10× a year

how fast the price of the same AI capability keeps falling

Autodesk–MaintainX release · Viture/NVIDIA release · Meta Wearables toolkit 1.0 · Microsoft Dynamics 365 retirement notice · a16z on LLM inference prices

COMPETITORS

Others show the steps or watch one station. Praxis watches the work, wherever it moves

	FOLLOWS THE PERSON	WATCHES THE WORK
	Praxis reads the shop's own manuals; on a phone today	
MaintainX · Augmentir steps on a phone; sign-offs, not sight	**Strivr** films and trains per customer; early access	
Dynamics 365 Guides Microsoft ends it 31 Dec 2026	**Viture + NVIDIA** sells the glasses; ships Q1 2027	
Manuals + videos explain the job; can't see you		
SHOWS THE STEPS		
Tulip · Dozuki apps at one station on the line		**Retrocausal · Invisible AI** fixed cameras at one workstation
		FIXED STATION

WHY WE WIN

No new hardware. A phone or tablet today, and the camera glasses shops are already starting to buy next.

No filming each customer. We compile the manuals and checklists a shop already has: 8 procedures for $4.16.

Proof, not a checkbox. We track the step, verify it and keep the record. We report our accuracy; none of them publish theirs.

Public product pages, checked 17 Oct 2026 · Viture Helix release, 16 Jun 2026 · UC Today review of Strivr, 13 Jul 2026 · detail in appendix A8

THE PLATFORMS

Meta, Google and Apple are building our hardware. A maintenance coach isn't their business

15M+

AI glasses forecast to ship in 2026. Each one is a camera we can run on, and we spend $0 building it.

Sept 2026

Meta opened its glasses' camera to outside apps. The platforms want apps like ours on top.

31 Dec 2026

Microsoft shuts down Dynamics 365 Guides. Big Tech tried frontline maintenance and is walking away.

THEY BUILD

Glasses, cameras and one general assistant for everyone. It answers questions about a single frame.

WE BUILD

The coach for one maintenance team at a time: its own procedures, step-by-step state, verified records and a pilot on site.

Omdia AI-glasses forecast · Meta Wearables Device Access Toolkit 1.0 · Microsoft Dynamics 365 retirement notice · Apple glasses reported for 2027

MARKET · US, PER YEAR, BOTTOM-UP

A $306M wedge in a $1.8B first market, in a $9B one

TAM — **$9.2B**

15.4M US workers who follow written procedures, at $50 a month for verified records

SAM — **$1.8B**

2.47M US maintenance and repair techs: coaching new hires + records for everyone

WEDGE — **$306M**

~255k new maintenance techs a year, coached for 6 months at $200 a month

YEAR 3 — **$3M ARR**

1% of new techs: ~2,550 trainees, roughly 100 maintenance teams

BLS OEWS national tables, May 2025 · prices are our hypotheses · year 3 is a goal, not a forecast · math in appendix A4

SAVINGS, PRICE + GO-TO-MARKET

Each new tech costs a shop $11–23k of senior time. We charge $1,200. One saved hour a week pays for it

$11–23k

of senior-tech time per new hire today, by our estimate

$1,200

per trainee: $200 a month for six months, after a short paid pilot

1 hour

a week of saved senior time pays for it, out of the 10–20 hours a week a senior tech spends on each new hire

FIRST 10

Founder-led paid pilots: facilities teams, service shops, and Dynamics 365 Guides customers who need a replacement

NEXT 100

Case studies, plus reading procedures straight from the maintenance apps shops already use

THEN 1,000

Distributors and glasses partners. Land one site, then every site, then records for the whole team

OUTREACH SO FAR: 14 EMAILS SENT 17 OCT · 56 MORE SCHEDULED FOR 19 OCT

Pricing is our hypothesis, anchored on Tulip's $100–250 per station per month · senior-time split is our assumption (appendix A5)

THE LONG GAME

Every coaching session builds data nobody else has. That data is how we climb from technicians to robots

NOW

Coach new maintenance techs

Paid by maintenance managers, per trainee

GATE: PAID PILOTS HIT THEIR METRIC

NEXT

The record of verified work

Paid by quality and audit teams; then field service and assembly

GATE: CUSTOMERS EXPAND WITHOUT US

THEN

Our own models

On the device, replacing the rented model: better margins

GATE: MATCH IT AT 1/10 THE COST

LATER

Train and check robots

Paid by robot labs, which already spend >$100M a year on real-world data

GATE: A PAYING ROBOT PARTNER

THE ASSET UNDER EVERY STEP

First-person video of skilled work, lined up to the procedure, with verified steps, real mistakes and the fixes that worked. Customers pay us to collect it, and our step models already improve with every recording (31% → 51%).

Robot-data spend: Micro1 CEO estimate · gates are conditions, not dates · learning curve on held-out public recordings (appendix A6)

THE ASK

$10k clears the bottleneck and gets a first paid maintenance pilot by April

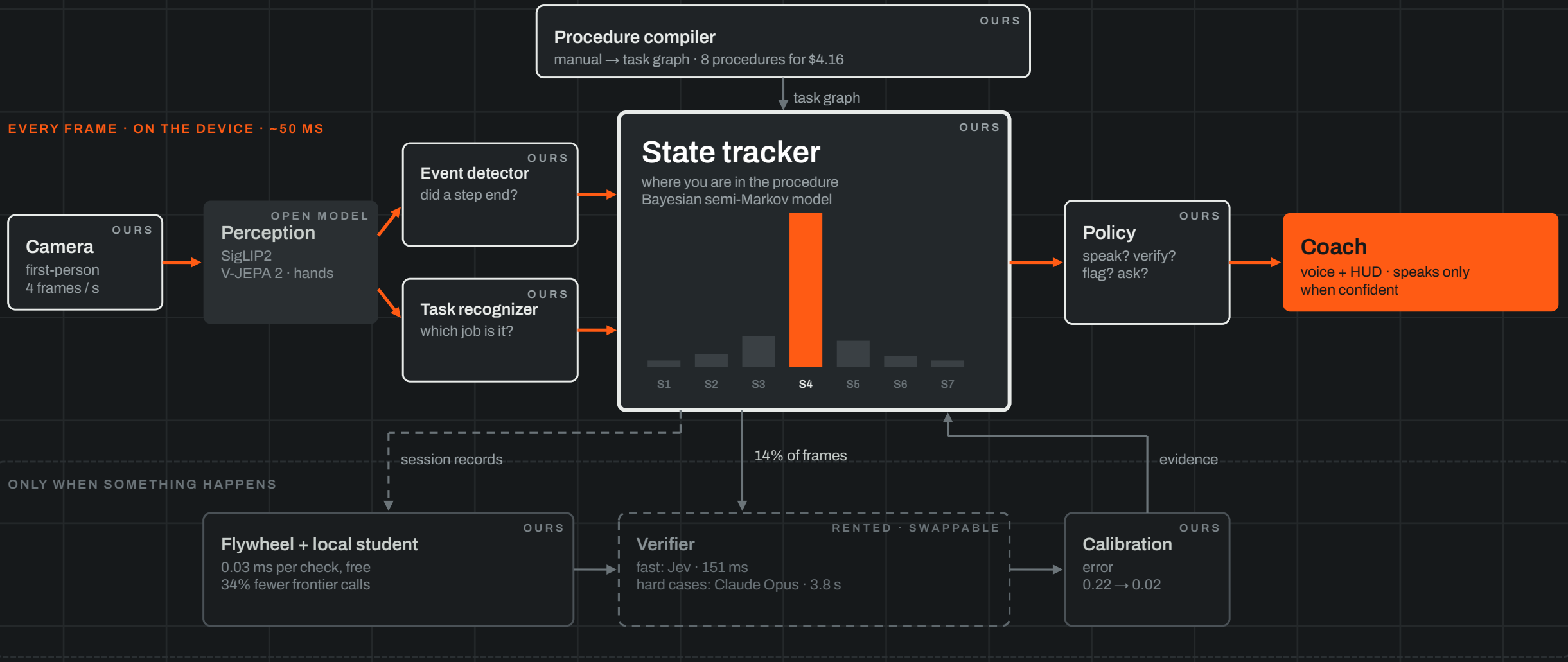
THE GAP TODAY	WHAT THE MONEY BUYS	AMOUNT	TRUE BY APRIL
Every number is from cooking video	Step-labeled recordings of real maintenance jobs	**$2,000**	**Measured on real maintenance footage**
Mistake alerts: 4 of 17 named a real error	Retraining the step and mistake models on that footage	**$3,000**	**Mistake detection re-measured on maintenance**
Runs on a laptop	Camera glasses and a capture kit	**$2,500**	**Running on glasses at a real site**
Live coaching costs money every hour	About 1,400 hours of pilot coaching, at ~$1.08 an hour	**$1,500**	**About 30 trainee-months of coaching covered**
No company a customer can sign with	Company formation and a pilot contract	**$1,000**	**A first paid pilot, with a metric agreed up front**

PILOT METRIC: TIME UNTIL A NEW TECH WORKS ALONE, AGREED WITH THE CUSTOMER BEFORE WE START

monishsaravana@college.harvard.edu · $10,000 over six months · same line items and amounts as the application

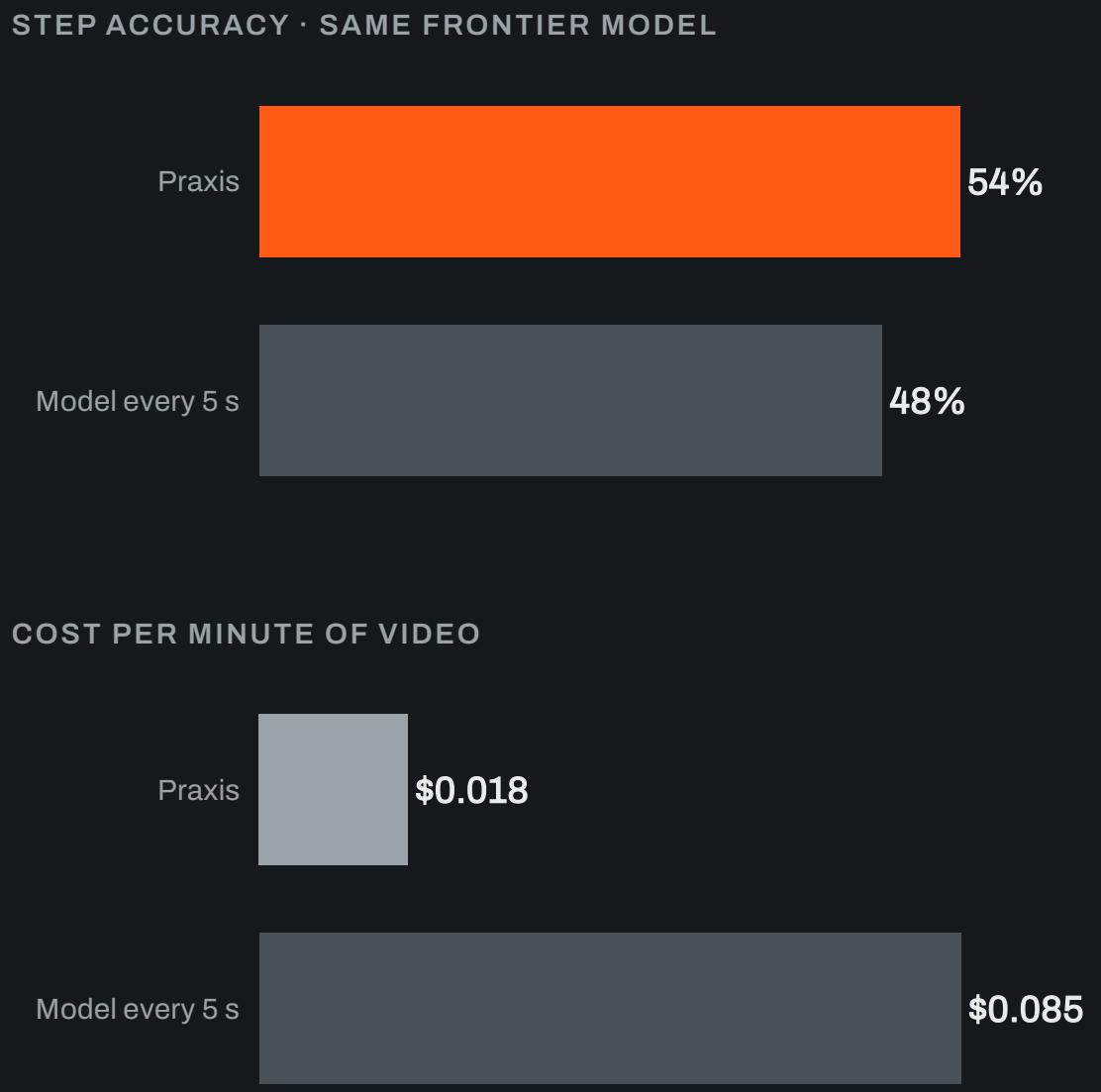
APPENDIX · A1

The frontier model is one swappable part. The tracking engine around it is ours.



Ours · Open model, on device · Rented, swappable

48 held-out clips of public first-person video · one run each



APPENDIX · A2

Same frontier model, with our tracking: 54% vs 48% step accuracy at a fifth of the cost

1/5

the cost per minute of video, and more accurate.

48 held-out public clips · one run each

The status quo: what each tool does today, and what it misses

SHADOWING

A senior tech, side by side

The real teacher today

MISSES About $44 an hour of senior time, one trainee at a time

MANUALS + VIDEOS

PDFs, binders, YouTube

Explain what should happen

MISSES Can't see what you're actually doing

PROCEDURE APPS

MaintainX, Tulip, Augmentir

Put the steps on a phone; collect sign-offs and photos

MISSES A checkbox isn't proof the step was done right

FIXED CAMERAS

Retrocausal, Invisible AI

Watch one workstation on a line

MISSES Maintenance moves around the machine

PRAXIS

Watches the work, follows the person around the machine, and runs on hardware a shop already owns.

Competitor pages checked 2026-10-17

APPENDIX · A4

Market math, and the assumptions behind it

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM	1% of new maintenance techs ≈ 2,550 trainees a year, ~100 mid-size teams	2,550 × $200 × 6 months	≈ $3M ARR

Prices are hypotheses. $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

BLS OEWS national tables, May 2025 · BLS Occupational Outlook Handbook

APPENDIX · A5

Unit economics, and the pricing risk

DESIGN	$ / VIDEO-MIN	$ / TRAINEE-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

THE RISK

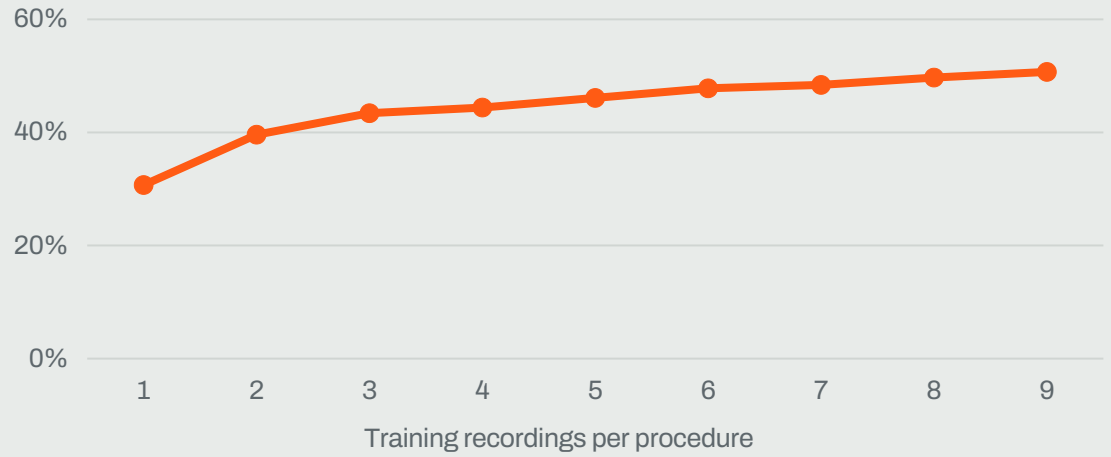
All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

2,520 coached minutes a month = 2 h/day × 21 days · senior-tech cost: BLS

Both learning curves: step models and the local student

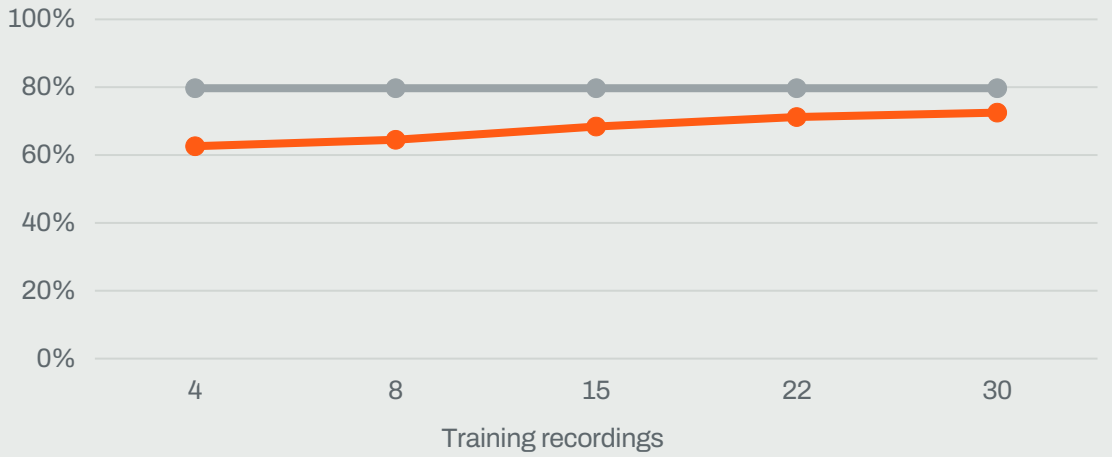
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

checking a step is done · grey: the frontier model, 79.7%



TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

Random subsets of public train-split recordings, 5 seeds per point

Training data, stated plainly

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

APPENDIX · A8

Who else is here, and how we differ

CATEGORY	EXAMPLES	US VS THEM
Instruction platforms	Tulip, Augmentir, Dozuki	They show the steps; we watch the work. Natural integration partners
Line cameras	Retrocausal, Invisible AI, Tulip Vision	Fixed at a station on a production line. Can't follow a person around a machine
First-person coaching	Viture Helix + NVIDIA, Strivr	Closest. Hardware-led or high-touch; no published accuracy
Toolkits	NVIDIA XR AI, Meta's camera toolkit, Android XR	Complements: the plumbing we build on
General assistants	Meta AI, Gemini Live	Answer questions about a frame. No plan, no state, no calibration

If a big platform ships a coach: they're building our hardware and plumbing. We go deep in one vertical, with its procedures, records and data.

Public product pages, checked 17 Oct 2026

Why not consumers first? Because the budget and the repeat data are in business

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

Autodesk–MaintainX release (28 May 2026) · Frontdoor pricing 2023–2024

The risks we see, and what we do about each

RISK	WHAT WE DO
Meta or Google ships a coach	They're building our hardware and plumbing. We go deep in maintenance with its procedures, records and data, and run on their glasses
Glasses aren't ready	It works on a tablet or phone today. Glasses are one more input adapter
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Privacy, unions, monitoring laws	Fast layers run locally; only event keyframes leave (14% of frames). Coaching for the worker, aggregates for the employer. Blur next
A wrong prompt on a safety-critical step	Lockout and similar steps stay the tech's job. Praxis prompts and records them, and never approves them
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the pilot

Glasses and voice: what's built, and what's designed

BUILT

- Runs live in real time on a recorded stream, with the local layers at about 50 ms per frame
- Model calls run in the background, so the live loop never waits (worst lag 0.8 s)
- Speaks its instructions; an urgent line cuts the queue
- Praxis Studio: the engine in a browser, on a camera, an upload or a test clip

DESIGNED, NOT BUILT

- Glasses input adapters (Meta camera toolkit, Android XR), one adapter pattern per device
- Voice input in natural language: "skip that", "why?", "no, I'm still on the belt"
- Push-to-talk by default; a protection headset in loud plants
- Face and badge blur; a local-model tier for sensitive sites

We say "designed to run on" glasses until an adapter has run on that device

I've built real-time vision on messy video before, and I built this in a week

LipTrac

Real-time lip reading from ordinary video. 60,000+ frames, 12 testers, provisional patent filed.

7 years

of competitive robotics, coaching teammates without seeing what they see.

1 week

to build the engine, a 48-video eval and a live app, solo. First commit Oct 17.

Monish Saravana · Harvard College · solo founder. Also: path planning for oil-cleanup boats from aerial imagery.